MAT1375, Classwork3, Fall2026

Ch3. Functions via Graphs

1. The definition of a Relation:

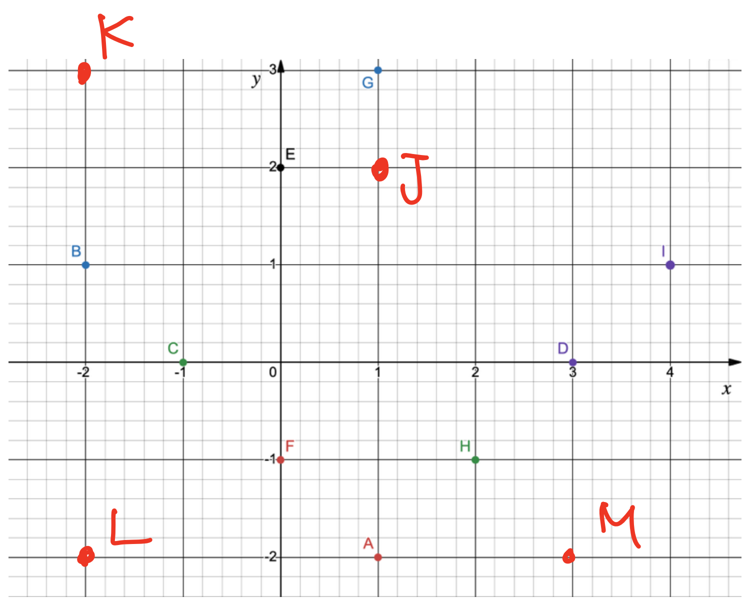
A ______________ is any set of ______________ pairs. The set of all first components of the ordered pair is called the ______________ of the relation and the set of all second components is called the ______________ of the relation.

2. Write down the ordered pair points in the given coordinate:

A: (1 , -2); B: (-2 , 1); C: (-1 , 0);

D: (3 , 0); E: (0 , 2); F: (0 , -1);

G: (1 , 3); H: (2 , -1); I : (4 , 1).



3. Plot the given points in a rectangular coordinate:

J:(1, 2); K:(-2, 3); L:(-2, -2); M:(3, -2).

4. From a relation to a function:

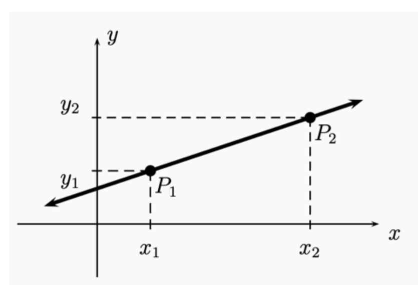
____.

5. Linear Functions and the Slopes:

A ______________ function is a function of the form

$$f(x) = \text{\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_}$$

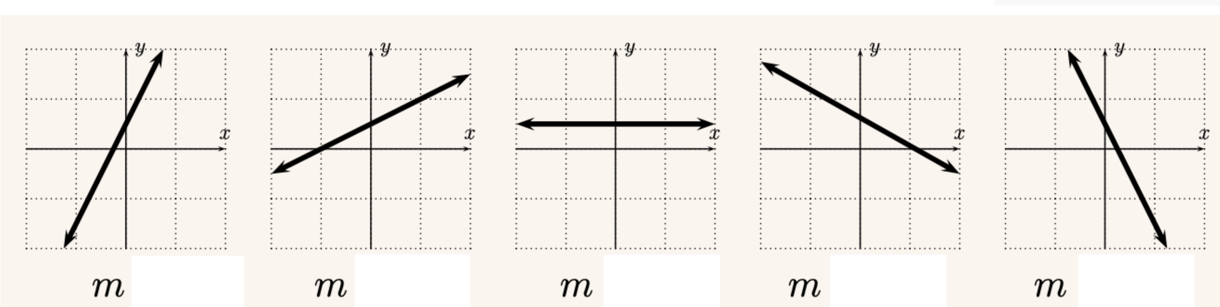
where m is the __________ of the line and $(0, b)$ is ______________ of the line. The domain of a line is ______________



6. The slope and its sign:

Given two points of a line $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$. Then the slope m is

$$m = \text{\_\_\_\_\_\_\_\_\_\_\_\_\_\_}$$



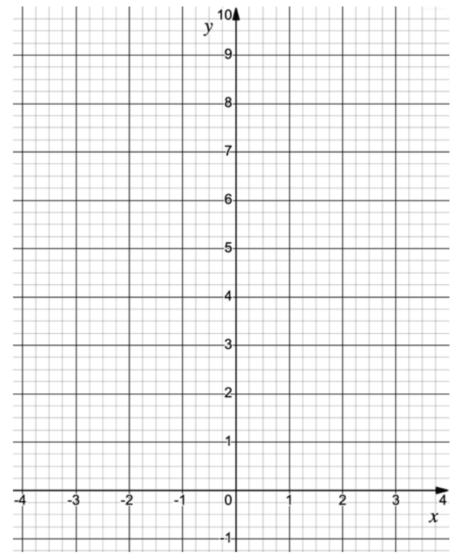
7. Find the equation of the line by the given information:

(1) $m = 2$ and y-intercept is $(0, -1)$

(2) the line passes $(1,1)$ and $(-1,-3)$

8. Functions given by graphs

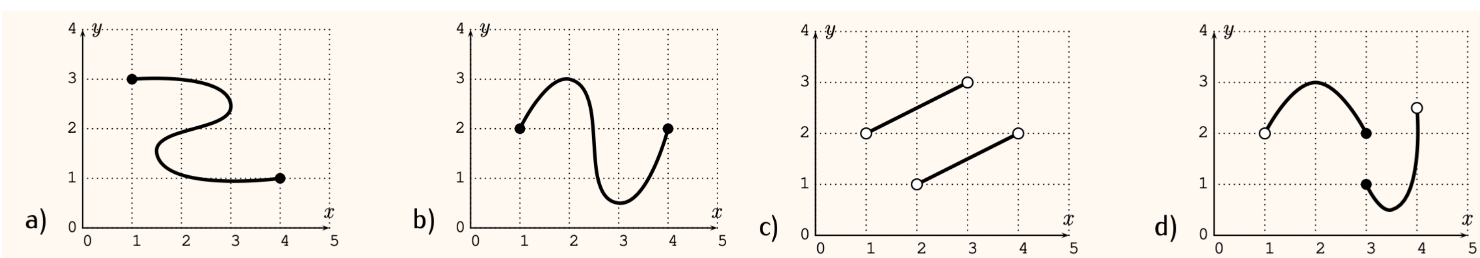
Let $y = x^2$ with domain $D = \mathbb{R}$. Graph this on the given coordinate.



9. Vertical Line Test:

A graph is the graph of a function precisely when every ______________ intersects the graph ______________.

10. Use **Vertical Line Test** to determine which of the following are the graphs of functions.



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11. Let f be the function given by the following graph.

(a) What is the domain of f?

(b) What is the range of f?

(c) For which x is $f(x) < 0$?

(d) Find $f(0) + 5$.

(e) Find $f(0 + 5)$.